TradeSight Export Analytics – Strategic Brief

Executive Briefing on Export Performance, Logistics SLA, Data Governance & Strategic Roadmap

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Production Dashboard: https://tradesight-export-analytics.vercel.app | **GitHub:** krupabt/TradeSight-Export-Analytics

1. Situation & Context

TradeSight Export Management operates as a simulated Chennai-based export management enterprise (modeled with ~45 operational personnel and 50 institutional clients across 12 destination countries). Prior to this initiative, commercial leadership lacked an integrated, auditable analytics platform to evaluate multi-year export revenue velocity, carrier delivery SLAs, physical return exceptions, and inventory capital allocation. To resolve this, the **TradeSight Export Analytics Dashboard V1** was engineered and deployed publicly on Vercel (**https://tradesight-export-analytics.vercel.app**) using React 18, Vite, and Recharts, providing seven specialized analytical modules: **(1) Executive Overview, (2) Export Revenue, (3) Logistics & Fulfillment, (4) Returns & Quality Audit, (5) Buyer Analytics & Segmentation, (6) Product Catalogue, and (7) Data Governance**.

All calculations originate strictly from the three original project datasets under zero-synthetic-data compliance: **da_b2b_orders.csv** (600 shipments, 13 columns; commercial wholesale orders), **da_product_catalogue.csv** (50 SKUs, 7 columns; master product specifications and margins), and **da_customer_profiles.csv** (200 accounts, 10 columns; direct retail consumer profiles).

Total Export Revenue **₹396.88 Cr** **600 Orders**	**Average Order Value** **₹66.15 Lakh** **1.51M Shipped Units**	**On-Time Delivery SLA** **77.50%** **135 Delayed Shipments**	**Avg Transit Time** **23.49 Days** **Median: 23.5 Days**	**Unit Return Rate** **1.01%** **15,242 Returned Units**	**Catalogue SKUs & Margin** **50 SKUs / 13,857 Stk** **58.62% Avg Markup**

2. Key Empirical Findings

Analytical Dimension	Empirical Evidence & Quantitative Findings (from Original CSV Datasets)
Commercial Revenue & Order Volume	Total export turnover reached **₹3,968,829,214.20 INR** (₹396.88 Cr) across **600 transactions**, with an AOV of **₹66.15 Lakh**. Revenue displays marked quarterly seasonality, peaking in Q4 2023 (₹65.73 Cr) and June 2024 (₹56.24 Cr).
Buyer & Supplier Structure	Transactions encompass **187 distinct commercial buyers** and **80 distinct suppliers/sellers**. Repeat buyers account for **227 orders (37.83%)**. Buyer volume is moderately concentrated, with the top 10 buyers generating approximately 15% of wholesale turnover.
Logistics SLA & Transit Times	Overall on-time delivery stands at **77.50% (465 orders)**, leaving **135 orders delayed (22.50%)**. Transit duration averages **23.49 days** (range: 10 to 35 days). Port dispatch delays and carrier bottlenecks account for major fulfillment variance.
Returns & Physical Anomalies	Total units returned were **15,242** out of 1,509,070 shipped (**1.01% return rate**). However, data audit reveals **two physical return overrun anomalies**: Order B2B-00131 (shipped 22, returned 28) and Order B2B-00188 (shipped 39, returned 48).
Master Catalogue & Inventory	Catalogue contains **50 SKUs** across 5 categories, with **13,857 warehouse stock units**. Average theoretical unit gross margin between retail Price_MRP and wholesale Price_Wholesale_COGS is **58.62%** (individual SKUs range from 46.15% to 66.67%).
Retail Consumer Profiles	The da_customer_profiles.csv dataset holds **200 retail consumers** generating **₹49.93 Lakh** with an overall **3.0% churn rate (6 accounts)**. 42 accounts exhibit chronological date inversions (First_Purchase_Date > Last_Purchase_Date).
Documented Dataset Limitations	Three requested enterprise metrics are mathematically unsupported by raw data and labeled **NOT AVAILABLE**: (1) **Foreign Exchange (FX) Impact** (all orders in INR), (2) **B2B Order Margin** (no SKU-level COGS in orders), and (3) **Customs Compliance** (no shipping bills).

3. Business Implications for Leadership Decision-Making

• **Fulfillment Bottlenecks Threaten Buyer Retention:** The 22.50% delay rate (135 orders) directly jeopardizes SLA compliance, risks contractual late delivery penalties, and erodes repeat purchase confidence among commercial buyers.
• **Data Quality Anomalies Reflect Upstream ERP Gaps:** Return overruns (returned units exceeding shipped units) and chronological date inversions demonstrate lack of validation controls at point of data entry, creating compliance vulnerabilities.
• **Catalogue Margins Must Not Be Confused with True Order Profitability:** The 58.62% catalogue markup represents retail reference pricing. B2B orders lack landed manufacturing costs, ocean freight, insurance, and export duties. Treating catalogue margin as actual commercial profit risks severe financial miscalculation.
• **Information Gaps Restrict Strategic Agility:** Sole reliance on INR denominations blinds leadership to foreign exchange fluctuations, while absence of customs tracking prevents operational bottleneck diagnostics at port terminals.

4. Three Prioritised Strategic Recommendations

Based on quantitative findings and business risk implications, the following three actionable, prioritized recommendations are presented for immediate operational adoption. In accordance with enterprise governance standards, no speculative percentage gains or unverified financial returns are claimed:

RECOMMENDATION 1: Improve Export Delivery Performance & Carrier SLA Governance	**PRIORITY: HIGH (IMMEDIATE)**
Identified Problem	135 export shipments (22.50%) were delayed past scheduled delivery dates, with an average transit time of 23.49 days and long-tail shipments extending up to 35 days, driving fulfillment uncertainty.
Proposed Action Plan	1. Audit all 135 delayed orders segmented by carrier, origin port, and freight mode. 2. Deploy end-to-end milestone tracking capturing port dispatch, customs clearance, ocean transit, and final delivery. 3. Establish a formalized 48-hour exception escalation review for any shipment exceeding 20 days in transit.
Expected Business Impact	Provides real-time shipment visibility, enables proactive carrier accountability, accelerates issue escalation prior to SLA breach, and strengthens long-term commercial buyer satisfaction and retention.
Key KPI to Monitor	**On-Time Delivery SLA % (baseline: 77.50%; target: >85.00%) | Delayed Shipment Count (baseline: 135 orders; target: <90 orders)**

RECOMMENDATION 2: Strengthen Upstream Data Quality & Validation Gates	**PRIORITY: HIGH (IMMEDIATE)**
Identified Problem	Critical data integrity exceptions were detected: two physical return overruns (B2B-00131: 28 returned vs 22 shipped; B2B-00188: 48 returned vs 39 shipped) and 42 customer profile chronological date inversions.
Proposed Action Plan	1. Audit upstream ERP/warehouse entry points for B2B-00131 and B2B-00188 to determine clerical vs. physical causes. 2. Implement strict hard input validation in ERP: reject return entries where return_qty > quantity_units without authorized management credit override. 3. Add database constraints enforcing Last_Purchase_Date >= First_Purchase_Date across CRM systems.
Expected Business Impact	Ensures 100% audit-compliant reporting, prevents distorted inventory asset valuations, eliminates physical impossibilities from commercial ledgers, and restores stakeholder confidence in analytical reporting.
Key KPI to Monitor	**Return Overrun Anomaly Count (baseline: 2; target: 0) | Chronological Date Inversion Count (baseline: 42; target: 0)**

RECOMMENDATION 3: Enhance Export Data Architecture & Financial Visibility	**PRIORITY: MEDIUM (STRATEGIC)**
Identified Problem	Existing data architecture lacks relational linkage between B2B orders and product SKUs, records transactions solely in INR without FX metadata, and omits customs/ICEGATE documentation attributes.
Proposed Action Plan	1. Implement relational order_line_items schema connecting order_id to Product_ID, capturing landed unit production costs, freight charges, and insurance. 2. Enrich transactions with foreign currency attributes (currency_code, booking_spot_rate, realized_rate). 3. Integrate customs clearance milestones (Shipping Bill number, Port of Loading, export duty clearance).
Expected Business Impact	Enables mathematically rigorous order-level gross and net profit margin analysis, empowers leadership to hedge against foreign exchange volatility, and ensures end-to-end export regulatory compliance.
Key KPI to Monitor	**SKU-to-Order Line-Item Linkage Rate (target: 100%) | True Net Margin Calculation Coverage (target: 100%)**

5. Conclusion & Operational Next Steps

Executive Conclusion: The TradeSight Export Analytics Dashboard V1 transforms raw export transactional data into an auditable, interactive decision-support engine. By providing real-time visibility across ₹396.88 Cr in export turnover, establishing rigorous data quality quarantines, and transparently isolating unsupported business metrics, the platform enables leadership to make evidence-based operational decisions.

Immediate Next Steps:
1. *Operational Review:* Convene carrier logistics review targeting the 135 delayed shipments identified in Section 3.
2. *ERP Engineering:* Deploy validation constraints in warehouse intake to eradicate return overruns and date inversions.
3. *Architecture Sprint:* Initiate data warehouse schema migration to introduce line-item SKU linkages and foreign currency capture.

Deliverable Links & Verification:
- **Live Production Dashboard:** https://tradesight-export-analytics.vercel.app
- **GitHub Project Repository:** https://github.com/krupabt/TradeSight-Export-Analytics
- **Prepared by:** Krupa B T | **Milestone:** Cynaris Data Analyst Internship — Final Dashboard + Strategic Brief